Hybrid Wheel-leg Locomotion in Rough Terrain



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Inspiration

- Human body is optimized for bipedal locomotion. But what happens if we encounter a complex terrain like confined space, water or soft terrain?



Fig 1(a): Crawling



Fig 1(b): Swimming



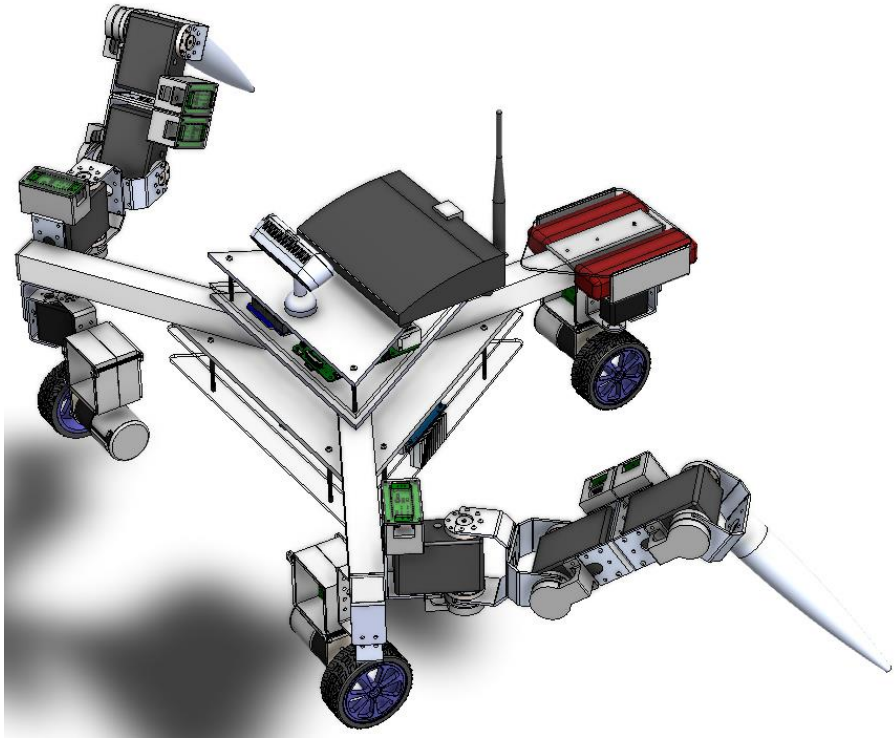
Fig 1(c): Skiing

Examples where we require more **points of contacts** as well as more **actuation.**

Abstract

- Wheeled robots can achieve high speeds with low power consumption, yet its ability to overcome obstacles, stability and ease of navigation in unstructured environments are generally limited
- Legged robots are suitable for applications in both structured and unstructured environments and they can overcome uneven terrains by adapting their gaits accordingly.
- However, legged robots are relatively slow, inefficient in energy consumption and require more actuators and complex control systems.
- We introduce a novel locomotion strategy of combining wheeled and legged actuation in a parallel structure.

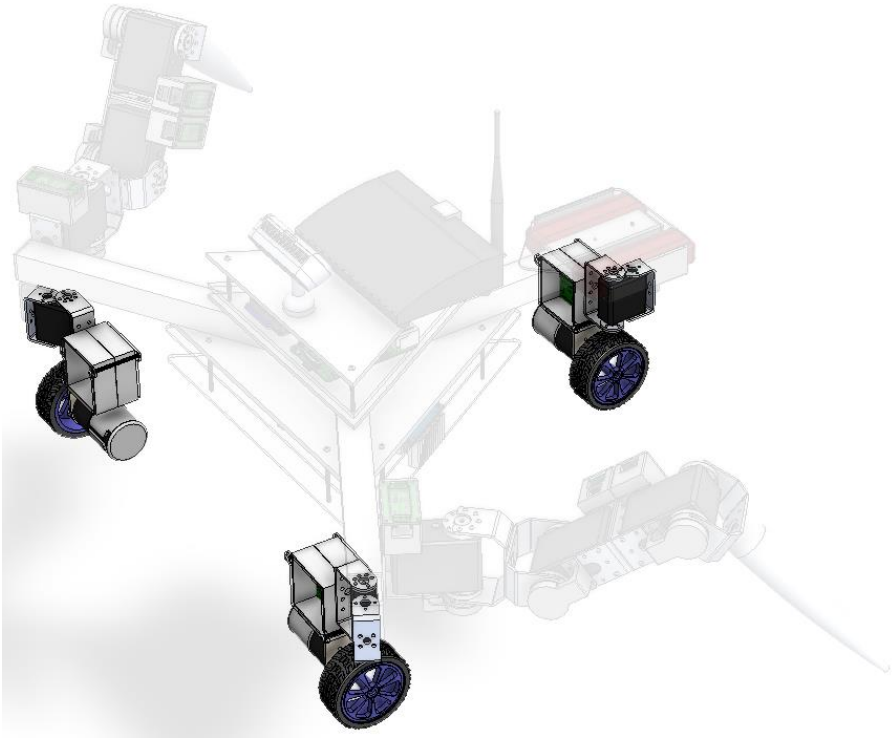
Prototype Design



- We present a prototype robot architecture to embody the concept and show experimentally the robot traversing a variety of indoor and outdoor terrains.

Fig 2(a): Prototype CAD design

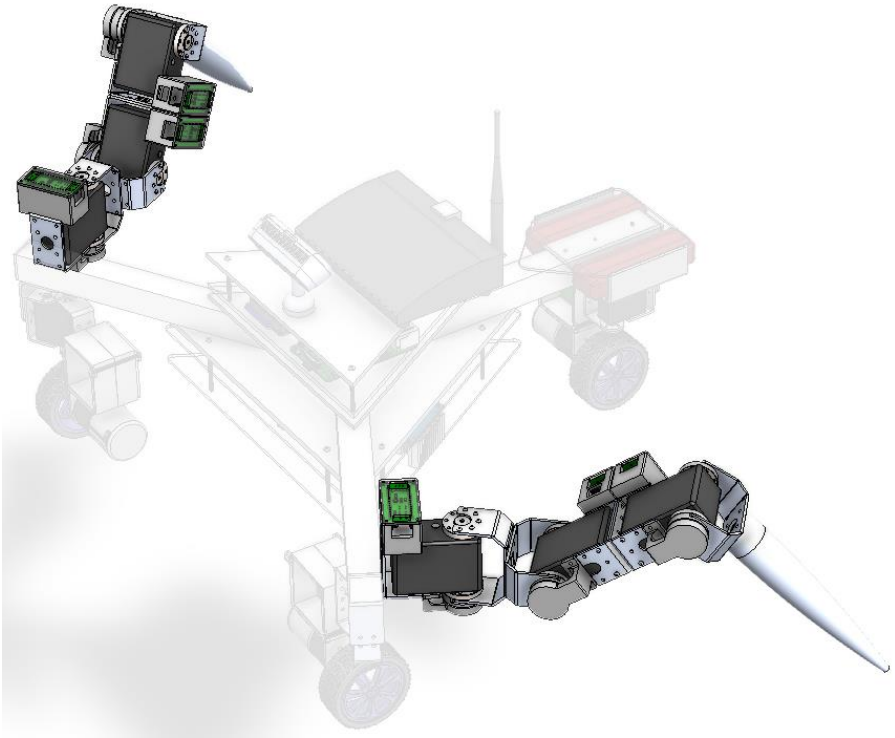
Wheeled Sub-system



- Contains three wheel modules with independent steering and propulsion
- Every module comprises sufficient low-level processing power to perform module specific commanding, sensing, and information feeding-back

Fig 2(b): Wheeled sub-system highlighted

Legged Sub-system



- Contains two leg modules with three degrees of freedom for each.
- Will be supported by the back wheel module for balancing and propulsion requirements in a unique wheel-leg crawling locomotion configuration.

Fig 2(c): Legged sub-system highlighted

Butterfly Gait

- A novel gait pattern called “buttery gait” is introduced for the robot that enables stable and efficient traversal of challenging terrains.

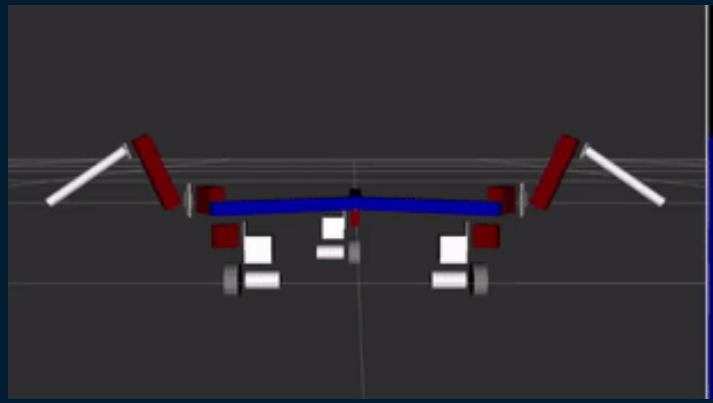


Fig 3(a): RViz visualitsation
(Trajectory is not in real-time)

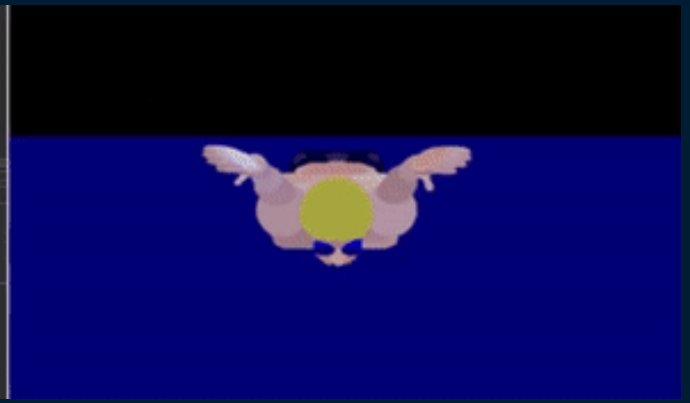


Fig 3(b): Butterfly stroke swimming
(Image source: Wikipedia)

Butterfly Gait

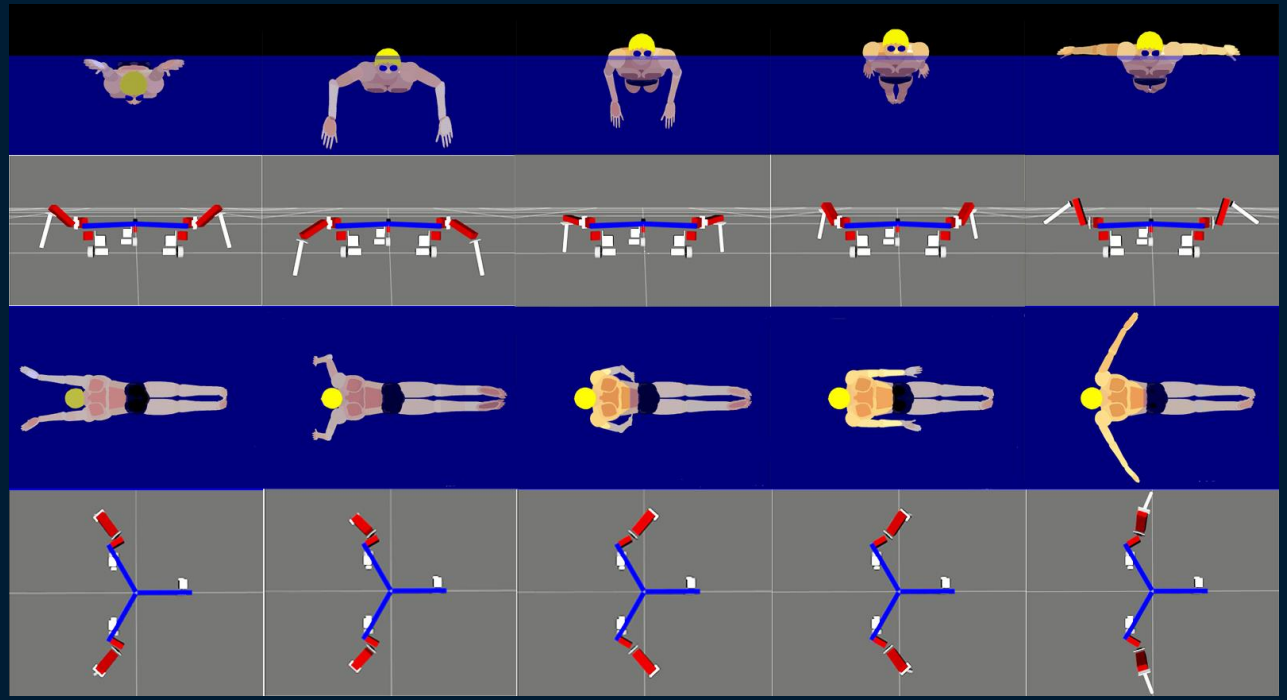


Fig 4: Correspondence between butterfly stroke swimming and butterfly gait

Variations of Butterfly Gait

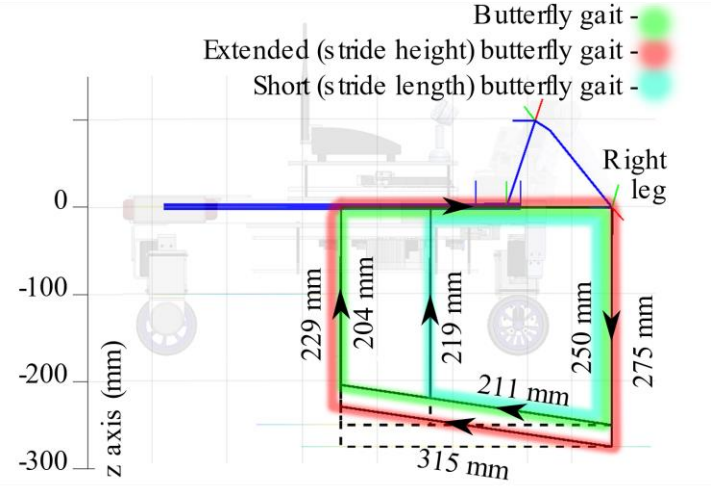


Fig 5(a): Right view

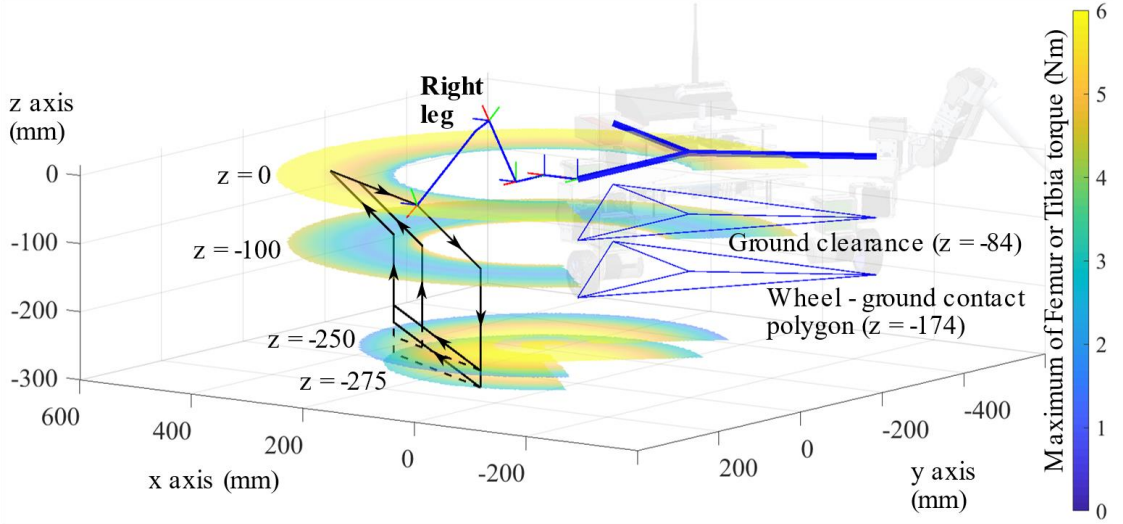
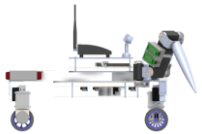


Fig 5(b): Isometric view





Autonomous Reconfigurability

- The robot dynamically reconfigures itself and chooses the suitable locomotion technique based on immediate terrain.
- Classification of the immediate terrain is performed using bag of visual words (BoVW) method from the real-time RGB video feed of the camera.
- Characterisation of the immediate terrain is performed based on a digital elevation model (DEM) created using the point cloud data feed from the camera and orientation information feed from inertial measurement unit (IMU).
- Root mean square height (RMSH), is used as the measure of surface roughness.

$$RMSH = \sqrt{\frac{\sum_{i=1}^{n}(z_i - \bar{z})^2}{n-1}}$$

n : number of data points,
z_i : is the elevation of the ith data point
$\bar{z}$: is the mean elevation of all n data points.

Identifying the Immediate Terrain

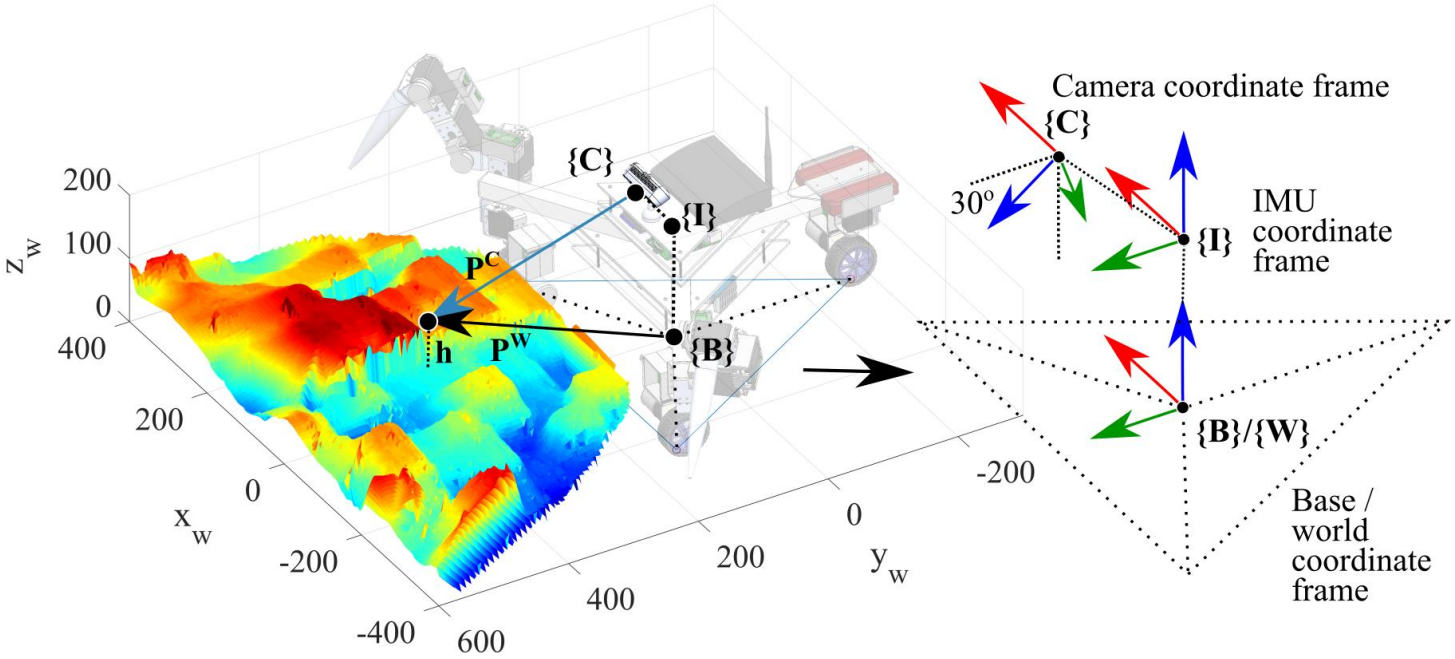


Fig 6: Coordinate frame assignment for creating the DEM

Identifying the Immediate Terrain

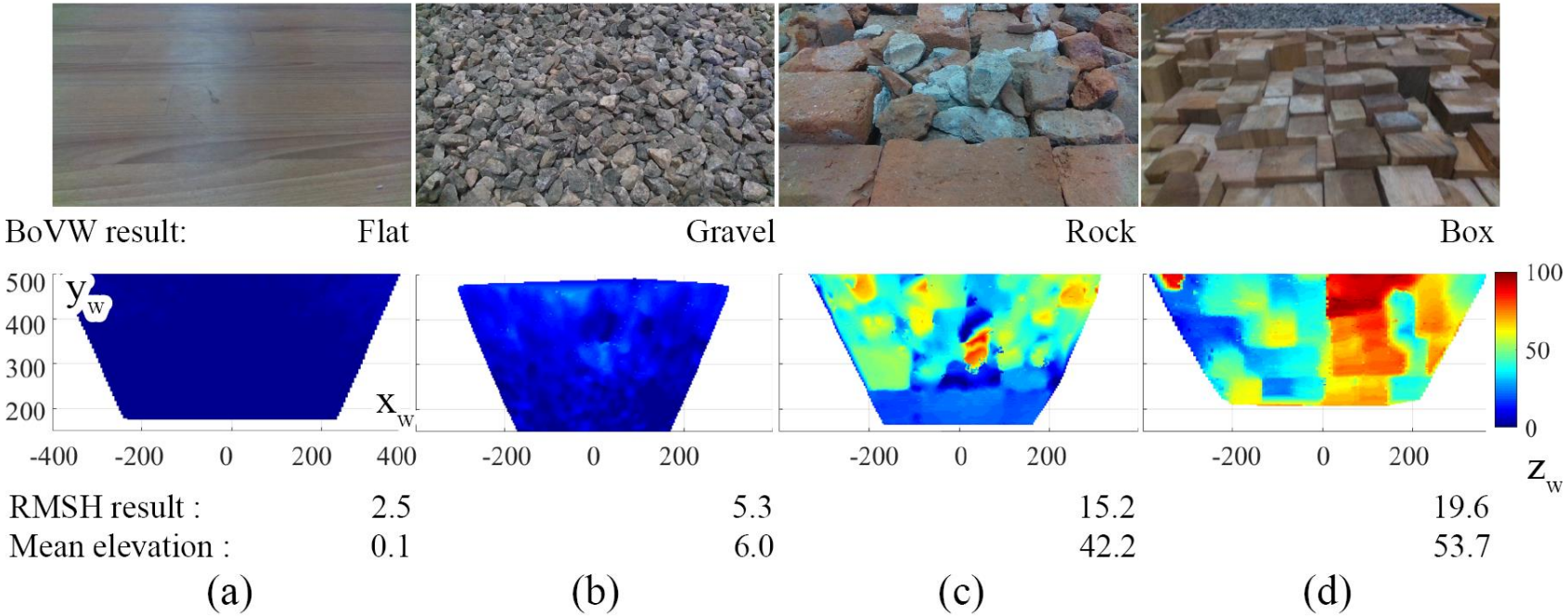


Fig 7: Classification and characterization results for indoor test bed

Hybrid Locomotion Selector

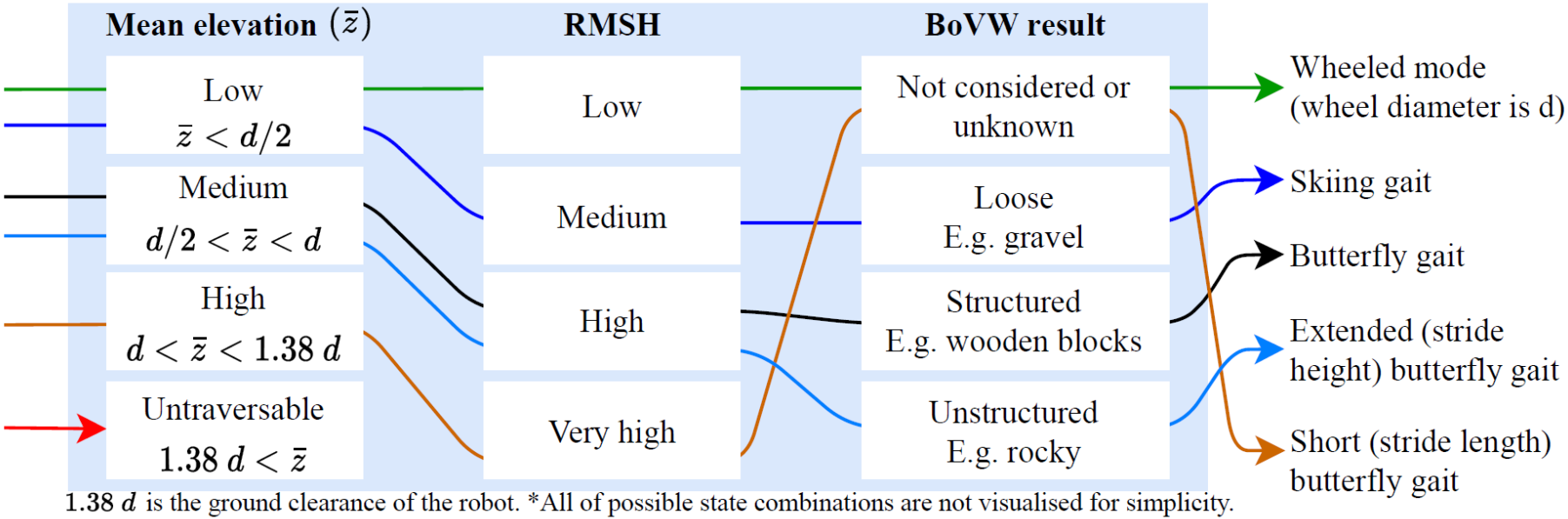


Fig 8: Primary states in hybrid locomotion selection node

Wheeled Sub-system

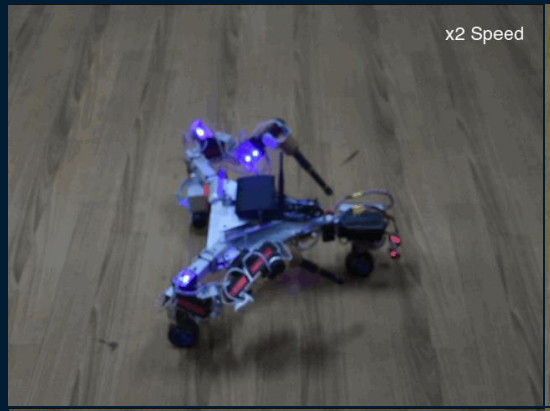


Fig 9(a)
Forwards &
Backwards motion

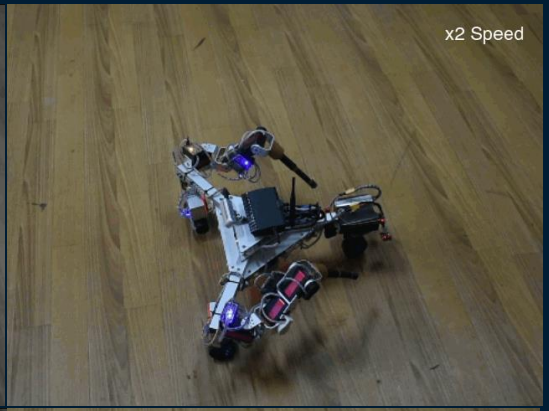


Fig 9(b)
Rotation
about yaw axis

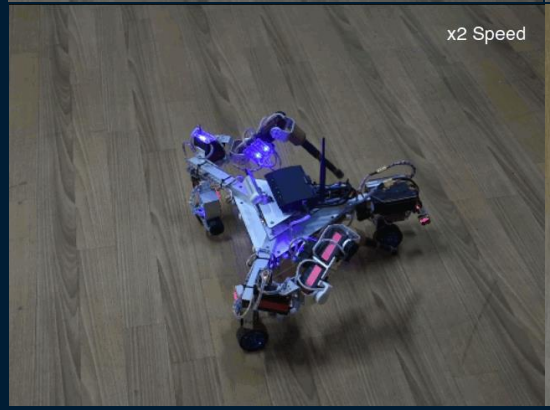


Fig 9(c)
Diagonal motion

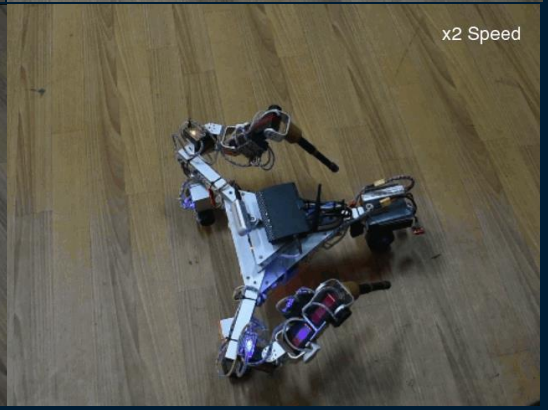


Fig 9(d)
Sideways motion

Obstacle Traversability

- The robot is capable of surmounting an obstacle of height 1:38 times the diameter of a wheel, which is also the ground clearance of the developed hardware.

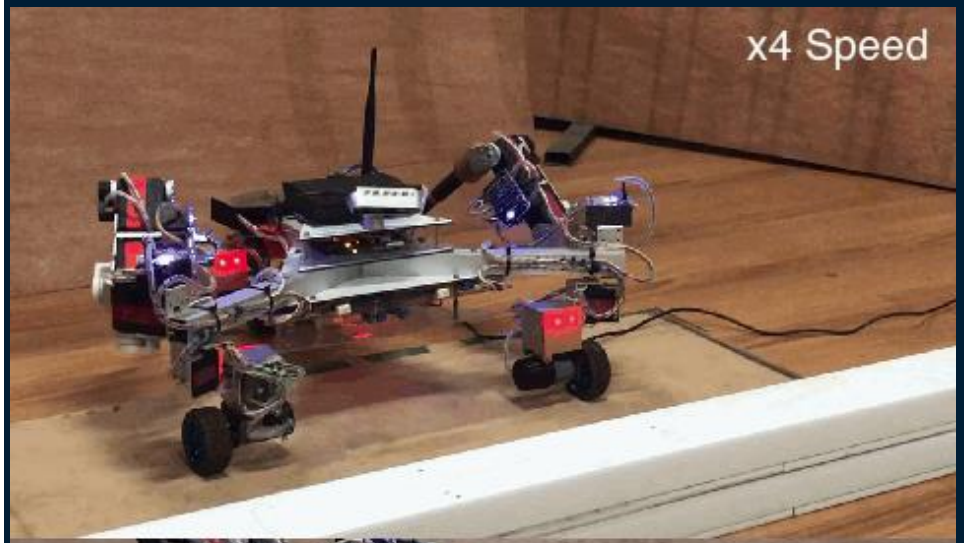


Fig 10(a):

Isometric view



Fig 10(b):

Right view

Indoor Test Bed Results



Fig 11(a)
Structured-uneven
Butterfly gait



Fig 11(a)
Unstructured-even
Skiing gait



Fig 11(c)
Unstructured-uneven
Short-butterfly gait



Playback speed x5

Outdoor Results



Fig 12(a)
Tree roots
Speed x3



Fig 12(b)
Gravel
Speed x2



Fig 12(c)
Step obstacle
Speed x4



Fig 12(d)
Grassy slope
Speed x2

Energy Efficiency of the Robot

Parameter	Wheeled mode	Butterfly gait	Short butterfly gait
Average cost of transportation (CoT)	4.95	41.77	71.28
Total energy consumption per 1m (kJ/m)	0.20	1.87	3.16
Range for 80% battery capacity (m)	695.60	75.10	44.52

$$CoT = \frac{P}{mgv}$$

P : overall power consumption of the robot
m : mass of the robot, which is 5.72 kg
g : gravitational acceleration
v : average velocity of the robot

Conclusion

- Manoeuvrability in a wide range of environments while maximising the average energy efficiency in compatible environments
- Legs actively contribute in maintaining the CoG of the robot to provide stability as well as the maximum traction for the wheels or provide propulsion in ramps.
- Legged subsystem can be used for certain manipulation tasks such as actuation or interaction with objects.
- Modularity allows easy diagnosis and replacement capacity for an operator.
- Removing one of the leg modules can reduce the curb weight of the fully assembled robot by 14.93% when operating in a structured-even terrain while increasing the payload limits by the same margin.








Thank You

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